```
clc;
clear;
close all;
```

FIR FILTER PARAMETERS

```
b = [0.0625 0.2500 0.3750 0.2500 0.0625];
Fs = 1000;
N = 128;

x = [1 zeros(1,N-1)];
n = 0:N-1;
```

1. DIRECT REALIZATION

```
yd = filter(b,1,x);
```

2. CASCADE REALIZATION

```
z = roots(b);
g = b(1);
yc = g*x;

for k = 1:length(z)
    yc = filter([1 -z(k)],1,yc);
end

yc = real(yc);
```

3. SYMMETRY AND LINEAR PHASE

```
symErr = max(abs(b-fliplr(b)));
D = (length(b)-1)/2;

if symErr < 1e-12
    symmetry = 'Symmetric';
else
    symmetry = 'Not Symmetric';
end
```

4. FREQUENCY, PHASE AND GROUP DELAY

```
[H,w] = freqz(b,1,1024);
phase = unwrap(angle(H));
gd = grpdelay(b,1,1024);
```

5. NUMERICAL COMPARISON

```
err = max(abs(yd-yc));
dev = err/max(abs(yd))*100;
```

6. DISPLAY RESULTS

```
fprintf('\n=======================================\n');
fprintf('     FIR FILTER REALIZATION RESULTS\n');
fprintf('=======================================\n');

fprintf('\nFIR Coefficients:\n');
fprintf(' %.4f',b);
fprintf('\n\n');

fprintf('Sampling Frequency = %d Hz\n',Fs);
fprintf('Number of Samples = %d\n',N);
fprintf('Coefficient Symmetry = %s\n',symmetry);
fprintf('Symmetry Error = %.3e\n',symErr);
fprintf('Group Delay = %.2f samples\n',D);
fprintf('Maximum Error = %.3e\n',err);
fprintf('Percentage Deviation = %.3e %%\n',dev);
```

```
=======================================
     FIR FILTER REALIZATION RESULTS
=======================================

FIR Coefficients:
 0.0625 0.2500 0.3750 0.2500 0.0625

Sampling Frequency = 1000 Hz
Number of Samples = 128
Coefficient Symmetry = Symmetric
Symmetry Error = 0.000e+00
Group Delay = 2.00 samples
Maximum Error = 8.327e-16
Percentage Deviation = 2.220e-13 %
```

7. FIRST 10 IMPULSE RESPONSE VALUES

```
fprintf('\n=======================================\n');
fprintf(' FIRST 10 IMPULSE RESPONSE VALUES\n');
fprintf('=======================================\n');

fprintf('%5s %12s %12s\n','n','Direct','Cascade');

for k = 1:10
    fprintf('%5d %12.6f %12.6f\n', ...
        n(k),yd(k),yc(k));
end
```

```
=======================================
 FIRST 10 IMPULSE RESPONSE VALUES
=======================================
    n       Direct      Cascade
    0     0.062500     0.062500
    1     0.250000     0.250000
    2     0.375000     0.375000
    3     0.250000     0.250000
    4     0.062500     0.062500
    5     0.000000     0.000000
    6     0.000000     0.000000
    7     0.000000     0.000000
    8     0.000000     0.000000
    9     0.000000     0.000000
```

8. CASCADE REALIZATION ZEROS

```
fprintf('\n=======================================\n');
fprintf(' CASCADE REALIZATION ZEROS\n');
fprintf('=======================================\n');

disp(z);
```

```
=======================================
 CASCADE REALIZATION ZEROS
=======================================
  -1.0002 + 0.0000i
  -1.0000 + 0.0002i
  -1.0000 - 0.0002i
  -0.9998 + 0.0000i
```

9. OUTPUT PARAMETER TABLE

```
Parameter = {
    'Direct realization output'
    'Cascade realization output'
    'Impulse response'
    'Phase response'
    'Comparison of realizations'
};

Observation = {
    'Direct FIR filtering'
    'Cascade of first-order sections'
    'Equal to FIR coefficient sequence'
    'Linear phase'
    'Equivalent transfer-function behavior'
};

Value = {
```

```
    sprintf('[%.4f %.4f %.4f %.4f %.4f]',yd(1:5))
    sprintf('[%.4f %.4f %.4f %.4f %.4f]',yc(1:5))
    sprintf('[%.4f %.4f %.4f %.4f %.4f]',yd(1:5))
    sprintf('Group delay = %.2f samples',D)
    sprintf('Max error = %.3e',err)
};

Unit_Note = {
    'Amplitude'
    'Amplitude'
    'h[n]'
    'Samples'
    'Numerical precision'
};

Output_Table = table(Parameter,Observation,Value,Unit_Note);

fprintf('\n=======================================\n');
fprintf(' OUTPUT PARAMETER TABLE\n');
fprintf('=======================================\n');

disp(Output_Table);
```

```
=======================================
 OUTPUT PARAMETER TABLE
=======================================
              Parameter
Observation
Value                                Unit_Note
    ___________________________________
_____________________________________________
_____________________________________________    ________________________

    {'Direct realization output' }    {'Direct FIR
filtering'                  }    {'[0.0625 0.2500 0.3750 0.2500 0.0625]'}
{'Amplitude'          }
    {'Cascade realization output'}    {'Cascade of first-order
sections'      }    {'[0.0625 0.2500 0.3750 0.2500 0.0625]'}
{'Amplitude'          }
    {'Impulse response'          }    {'Equal to FIR coefficient
sequence'    }    {'[0.0625 0.2500 0.3750 0.2500 0.0625]'}
{'h[n]'               }
    {'Phase response'            }    {'Linear
phase'                         }    {'Group delay = 2.00
samples'          }    {'Samples'            }
    {'Comparison of realizations'}    {'Equivalent transfer-function
behavior'}    {'Max error = 8.327e-16'               }    {'Numerical
precision'}
```

GRAPH 1: DIRECT VS CASCADE IMPULSE RESPONSES

```
figure('Name','Direct and Cascade FIR Impulse Responses');

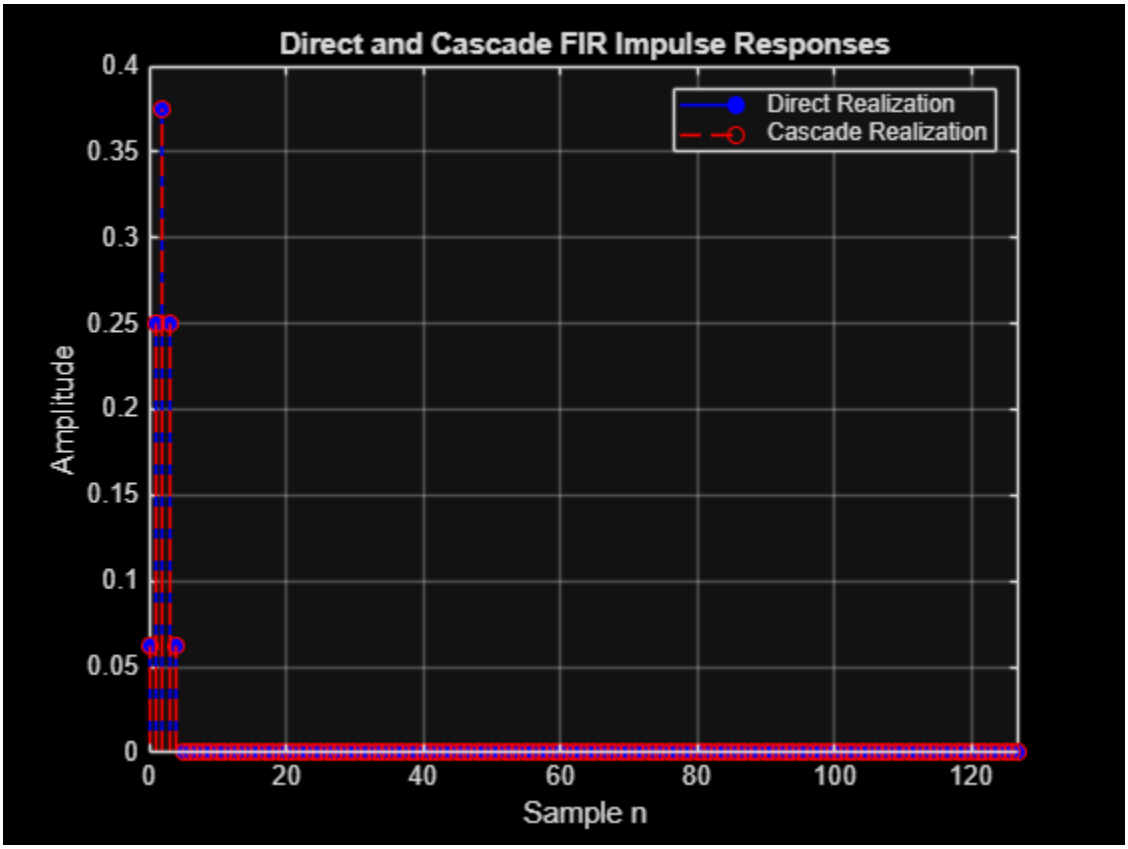
h1 = stem(n,yd,'b','filled');
hold on;
h2 = stem(n,yc,'r');

set(h2,'LineStyle','--','Marker','o');

grid on;
xlim([0 N-1]);

xlabel('Sample n');
ylabel('Amplitude');
title('Direct and Cascade FIR Impulse Responses');

legend('Direct Realization','Cascade Realization');
```



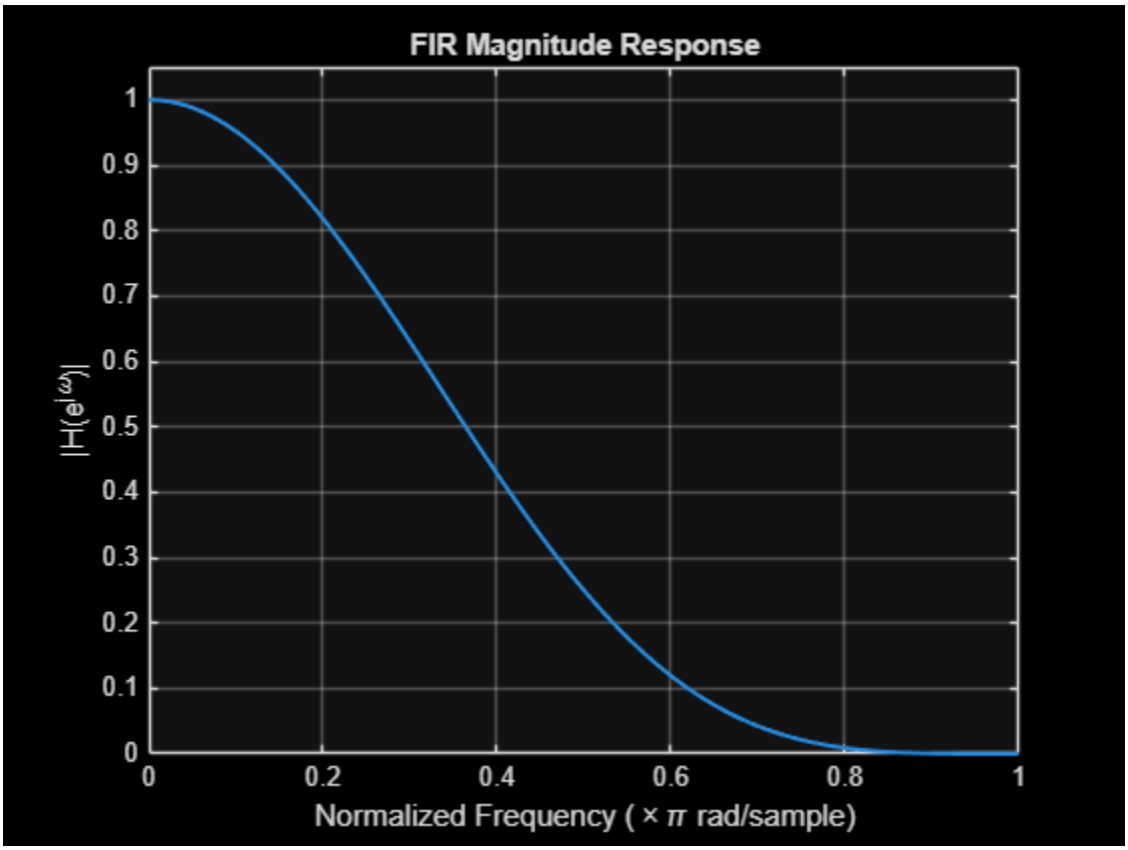
GRAPH 2: MAGNITUDE RESPONSE

```
figure('Name','FIR Magnitude Response');
```

```
plot(w/pi,abs(H),'LineWidth',1.5);

grid on;
xlim([0 1]);
ylim([0 1.05]);

xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('|H(e^{j\omega})|');
title('FIR Magnitude Response');
```



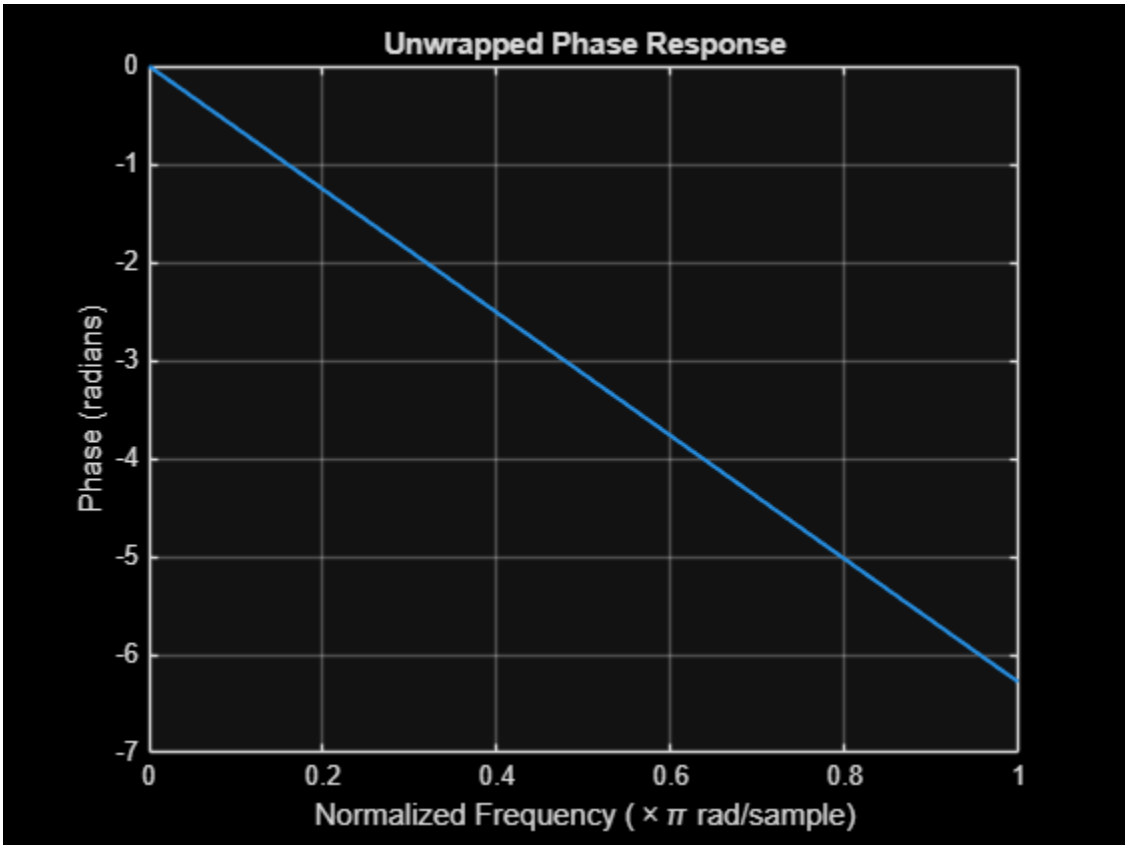
GRAPH 3: UNWRAPPED PHASE RESPONSE

```
figure('Name','Unwrapped Phase Response');

plot(w/pi,phase,'LineWidth',1.5);

grid on;
xlim([0 1]);

xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Phase (radians)');
title('Unwrapped Phase Response');
```



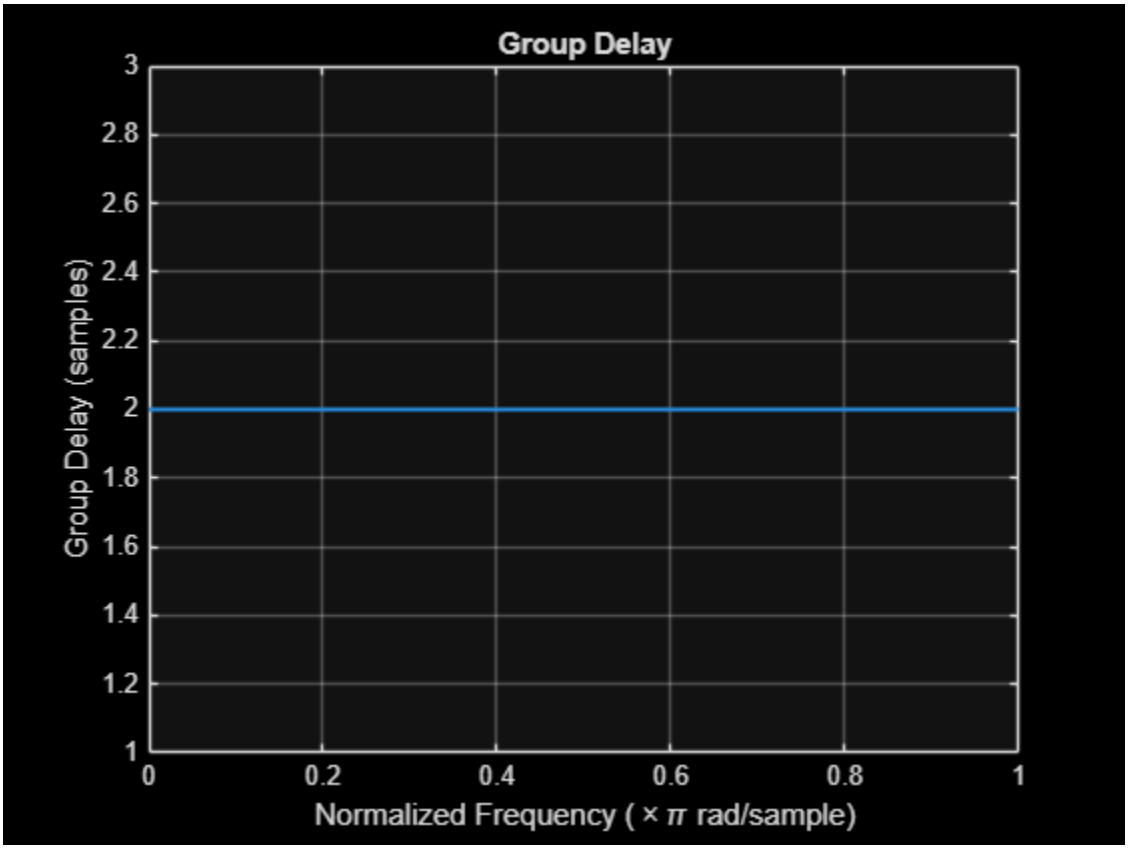
GRAPH 4: GROUP DELAY

```
figure('Name','Group Delay');

plot(w/pi,gd,'LineWidth',1.5);

grid on;
xlim([0 1]);
ylim([1 3]);

xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Group Delay (samples)');
title('Group Delay');
```



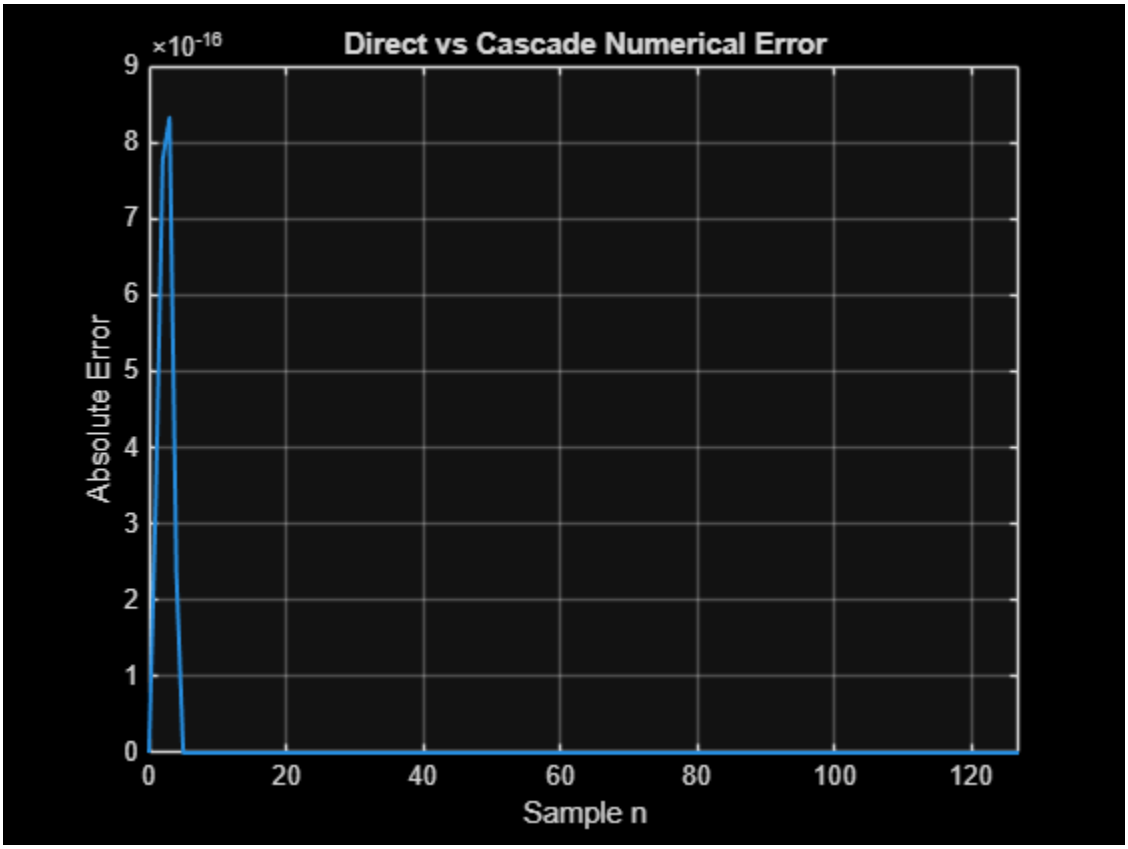
GRAPH 5: DIRECT VS CASCADE NUMERICAL ERROR

```
figure('Name','Direct vs Cascade Numerical Error');

plot(n,abs(yd-yc),'LineWidth',1.5);

grid on;
xlim([0 N-1]);

xlabel('Sample n');
ylabel('Absolute Error');
title('Direct vs Cascade Numerical Error');
```



10. FINAL RESULT

```
fprintf('\n=======================================\n');
fprintf(' FINAL RESULT\n');
fprintf('=======================================\n');

fprintf('1. FIR filter realized using Direct structure.\n');
fprintf('2. FIR filter realized using Cascade structure.\n');
fprintf('3. Coefficients are %s.\n',symmetry);
fprintf('4. Linear-phase group delay = %.2f samples.\n',D);
fprintf('5. Maximum realization error = %.3e.\n',err);
fprintf('6. Percentage deviation = %.3e %%.\n',dev);
fprintf('7. Direct and Cascade responses are equivalent.\n');
```

```
=======================================
 FINAL RESULT
=======================================
1. FIR filter realized using Direct structure.
2. FIR filter realized using Cascade structure.
3. Coefficients are Symmetric.
4. Linear-phase group delay = 2.00 samples.
5. Maximum realization error = 8.327e-16.
6. Percentage deviation = 2.220e-13 %.
7. Direct and Cascade responses are equivalent.
```

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